ordinary matter. Because the ordinary matter had pressure, and mixed with the radiation, at first only the dark matter was able to clump together due to gravity without immediately bouncing back. Later on, when the universe expanded more and the radiation streamed away from the cooling matter, gas was able to fall into these gravitational wells and begin to condense as stars and galaxies. Even today, the structure of matter on the largest scales —the cosmic web of galaxies and clusters of galaxies—is scaffolded by a network of dark matter clumps and filaments. At Cosmic Dawn, those invisible clumps and filaments first started to light up, as stars and galaxies ignited and shone, sparkling along the network like fairy lights in the darkness.

THE ERA OF GALAXIES

The next big transition for the universe came when so much starlight was coursing through space that it was able to ionize the ambient gas that had become neutral at the end of the cosmic fireball stage. This intense starlight broke hydrogen atoms back apart into free electrons and protons, creating giant bubbles of ionized hydrogen gas surrounding the brightest collections of galaxies. Those bubbles expanding through the cosmos marked the Epoch of Reionization (“re-” because the gas had been ionized during the Big Bang at the beginning, and was now being ionized again by the stars). That transition, which was completed at about the billion-year mark, is now one of the frontiers of observational astronomy, and we are only just beginning to understand how and when it occurred. In the almost 13 billion years since that time, things have carried on in much the same way, with galaxies forming and combining, supermassive black holes building up mass in galactic centers, and new stars being born and living out their lives.

So, here we are. The cosmos as we see it today is a vast, beautiful web of galaxies shining in the darkness. Our own pretty blue-and-white world orbits a moderately sized yellow star in a galaxy that is, in every meaningful way, fairly close to the average. While we have yet to find clear signals, this unremarkable galaxy might be teeming with life, as the debris of long-exploded supernovae creates the basic ingredients of biology on each of the worlds scattered around a hundred billion stars. By current estimates, as many as one in ten star systems has a planet of just the right